

Math 74: Algebraic Topology

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Problem 1. Compute the simplicial homology for $\mathbb{RP}^2$ using a Δ -complex structure on it.

Problem 2. Prove directly that singular homology is an invariant of homeomorphism class of the space. Here 'directly' means without referring to the fact that singular homology is an invariant of homotopy equivalence class of the space.

Problem 3. Compute the singular homology groups of a point.

Problem 4. Compute the homology groups of the Δ -complex X obtained from Δ^n by identifying all faces of the same dimension. Thus X has a single k simplex for each $k \leq n$.

Problem 5. Compute $H_i(D^n, S^{n-1})$ for all i and n .

Problem 6. Show that if A is a retract of X then the map $H_n(A) \rightarrow H_n(X)$ induced by the inclusion $A \subset X$ is injective. Now show that there is no retraction from D^n to S^{n-1} , for any n .

Problem 7. Let $f : (X, A) \rightarrow (Y, B)$ be a map such that both $f : X \rightarrow Y$ and the restriction $f : A \rightarrow B$ are homotopy equivalences.

1. Show that

$$f_* : H_n(X, A) \rightarrow H_n(Y, B)$$

is an isomorphism for all n .

2. For the case of the inclusion

$$f : (D^n, S^{n-1}) \hookrightarrow (D^n, D^n - \{0\}),$$

show that f is not a homotopy equivalence of pairs — there is no

$$g : (D^n, D^n - \{0\}) \rightarrow (D^n, S^{n-1})$$

such that fg and gf are homotopic to the identity through maps of pairs. (Observe that a homotopy equivalence of pairs $(X, A) \rightarrow (Y, B)$ is also a homotopy equivalence for the pairs obtained by replacing A and B by their closures.)

Problem 8. Show that chain homotopy of chain maps is an equivalence relation.